

NAPCORE Helpdesk: requirements and development

Presentation Draft

NAPCORE Helpdesk Team

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Problem And Goal

- ▶ Need a trusted answer layer for multimodal standards implementation.
- ▶ Main risk is fast but uncited guidance that creates delivery risk.
- ▶ Goal is practical answers with traceable references and controlled escalation.

Stakeholders And System Context (C4 Level 1)

Five stakeholder groups share one helpdesk interface:

- ▶ PTO (Public Transport Operator) — responsible for daily service delivery.
- ▶ PTA (Public Transport Authority) — sets policy and compliance standards.
- ▶ Developer — implements integrations and APIs.
- ▶ ITS System Integrator — connects helpdesk to broader mobility ecosystem.
- ▶ Ticketing Agent — manages passenger-facing systems.

External systems:

- ▶ Approved GitHub repositories (NeTEx-CEN/NeTEx) — knowledge source.
- ▶ LLM provider — optional grounded generation path.

C4 Level 1 - NAPCORE Helpdesk System Context



Asks standards
implementation
questions

Documentation Scope

- ▶ Standards: Transmodel, NeTEx, SIRI, OJP/OpRa, DATEX II.
- ▶ Priority source: NeTEx-CEN/NeTEx and approved companion repositories.
- ▶ Surfaces: separate user route (/user) and operator route (/operator) over one shared backend shell.

Core Approach

- ▶ FAQ-first with RAG fallback.¹
- ▶ FAQ-first: return approved answers when confidence is high.
- ▶ RAG fallback: retrieve source chunks, then generate a grounded answer.
- ▶ Generation profiles:
 - ▶ deterministic-grounded (active): question-adaptive deterministic templates with retrieval-score confidence.
 - ▶ llm-ready (optional): retrieved chunks passed to LLM synthesis when enabled.
- ▶ If evidence is weak, abstain explicitly.

¹RAG fallback means that when no high-confidence FAQ answer is available, the system retrieves relevant source chunks from approved repositories and generates an answer grounded in that retrieved evidence. In deterministic mode this is template-driven and score-based; in LLM-ready mode retrieved chunks are synthesized by the LLM while remaining evidence-bound.

Functional Requirements Highlights

- ▶ FR-001: question answering through the helpdesk UI.
- ▶ FR-004: ingestion and indexing of approved GitHub sources.
- ▶ FR-005: retrieval and grounded generation.
- ▶ FR-006: anti-hallucination and boundary enforcement.
- ▶ FR-002 and FR-003: editorial curation and FAQ promotion.

Current Product State

- ▶ Frontend now exposes separate routed surfaces for user chat and operator console.
- ▶ Backend supports deterministic grounded generation and an optional LLM-ready mode.
- ▶ Knowledge index is built from approved repositories and selected companion repositories.
- ▶ Shared connection state keeps auth and backend targeting consistent across both routes.

Current Chat UX

- ▶ User route and operator route are shown in full-size screenshots.
- ▶ Connection fields are intentionally blank for presentation clarity.

The screenshot displays a web application interface for NAPCORE HELPDESK. At the top, a dark teal header contains the text "NAPCORE HELPDESK" and "Q&A about NAPCORE related multimodal standardisation". Below this, a subtitle reads "Demonstration shell with separate user and operator surfaces sharing one backend connection.".

The main content area is divided into several sections. The first section is titled "Experience Switcher" and includes the text "Use separate routes for demo: user-facing chat and operator-facing console are now independently addressable." To the right of this text are two buttons: "User Chat" (highlighted in teal) and "Operator Console" (in a light grey box).

The second section is titled "Connection" and contains two input fields: "API Base URL" and "JWT Bearer Token". Both fields are currently empty. Below these fields is a checkbox labeled "auto-create dev JWT on page reload", which is checked.

The third section is titled "USER WORKSPACE" and "User Chat Interface". It includes the text "Independent user-facing conversation area (planned standalone surface)." and a smaller version of the NAPCORE HELPDESK header and Q&A title.

At the bottom of the visible area, the text "Web GUI chat for users" is visible.

Documentation And Delivery Artifacts

- ▶ Functional requirements, testing pack, and local run guide.
- ▶ RAG architecture and updated C4 architecture documents.
- ▶ Technology baseline: Django 5 + DRF backend, React/Vite frontend.
- ▶ Separate routed user/operator UX and deterministic grounded answering.
- ▶ PlantUML and exported SVG architecture assets for reproducible documentation.

Next Steps

1. Complete pilot testing across user chat, operator flow, and promotion workflow.
2. Evaluate deterministic mode outcomes against grounded LLM mode on the same question set.
3. Finalize repository ingestion profiles and production hardening backlog.

Appendix: Example Questions

Question 1: How to use NeTEx for exchanging a timetable?

NAPCORE HELPDESK

Q&A about NAPCORE related multimodal standardisation

Demonstration shell with separate user and operator surfaces sharing one backend connection.

Experience Switcher

Use separate routes for demo: user-facing chat and operator-facing console are now independently addressable.

User ChatOperator Console

Connection

API Base URL

/api/v1

☒ auto-create dev JWT on page reload

JWT Bearer Token

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJ1c2VybWxvY2FsaWwcmF

USER WORKSPACE

User Chat Interface

Independent user-facing conversation area (planned standalone surface).

User question

How to use NeTeX for exchanging a timetable?

view user chat for users

Question 2: How to implement IDs for a Stop Place registry?

Q&A about NAPCORE related multimodal standardisation